Part of Speech Tagging for English Text Using Second Order Markov Model

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Abstract

In natural language processing, one of the problem for sentence parsing is to assign the correct part of speech category to each word. This kind of task also known as tagging. In this paper, we present a tagging system based on 2nd order markov. First, we trained the system with a training data which was derived from a manually tagged LOB corpus. After that, we tested the system on our 116,785 words test data which was also derived from the LOB corpus. We carried several test. On its best performance, our system reached an accuracy of 95.586%.

1 Introduction

English vocabulary contains many ambiguous words. Ambiguous word has more than one part of speech categories. This kind of word appears in the sentence with different part of speech category. It depends on the context of that sentence. De Rose[2] has investigated the word ambiguity that occured in the Brown Corpus, the one million words corpus widely used on English text research. He found out that 11.5% of words on the corpus' vocabulary and 40% of running words are categorically ambiguous. Meanwhile, the part of speech category of a word reflects its syntactic and semantic role in a sentence. We cannot interpretate a word correctly without disambiguating the word – that is to assign the correct part of speech category to the word. This task is also known as tagging.

English text tagging is widely researched due to the need of text with disambiguated words. Disambiguated words in a document is useful for text analysis in information retrieval systems. Also, many language models for speech recognition systems are based on tagging model of word sequences.

Tagging systems usually use a dictionary containing vocabulary words and set of possible part of speech categories that each word in the vocabulary can assume. Klein et al.[5] used 30 categories on their Klein and Simmons algorithm. They claimed an accuracy of 90% in tagging the words. Greene et al.[4] were the first to attempt tagging a large sample of data. Using a palette of 86 part of speech categories, they succeeded in tagging correctly 77% of the million words in The Brown Corpus. Two systems mentioned above are rule-based. Another main approach commonly used in tagging system is probabilistic approach. In this approach, we need a large corpus to construct the dictionary containing word, part of speech category combinations and their probabilities. Charniak et al.[1] divided the Brown Corpus into two: training corpus and test corpus. The training corpus was used to construct the system's dictionary. The system was then tested on the test corpus. Some other example of achievement through

probabilistic approach are achieved by Kupiec[6] and Merialdo[7]. Kupiec introduced word equivalence classes as substitutes for words. By doing this, he succeeded in reducing the size of the dictionary.

All of the systems mentioned above use a dictionary that contains words and the list of possible part of speech category(ies) for each word. In this paper, we present a tagging system that uses not only the above dictionary, but also a new kind of dictionary contains statistic of all kind of words sequences or part of speech categories sequences that appear at least once somewhere in our 1,040,475 words training data. On its best performance, our system yielded an accuracy of 95.586% on a 116,785 words test data with 149 part of speech categories. The data was derived from the Lancaster-Oslo Bergen (LOB) Corpus which was already manually tagged.

2 Mathematical Model

First, let us assume that the English language that we know has some fixed vocabulary. We decide to make the mathematical expression for this vocabulary as $W = \{w^1, w^2, \ldots, w^x\}$ where $w^1, w^2, \ldots, w^x$ are the member words of the vocabulary, and x is the number of the words. For example, W could be like $W = \{a, aback, \ldots, zymotic\}$. Secondly, we assume that the English language also has some fixed part of speech categories. Like the vocabulary above, we make a mathematical expression for these categories, that is $T = \{t^1, t^2, \ldots, t^y\}$ e.g. $\{adjective, adverb, \ldots, verb\}$ where $t^1, t^2, \ldots, t^y$ are the part of speech categories and y is the number of the categories.

After that, we make assumption on sentences occured on English language. We assume that sentence is a words' sequence which could be expressed as $w_1, w_2, \ldots, w_n$ where n is the number of words on the sentence. Any word could become one of $w_1, w_2, \ldots, w_n$ as long as it is a member of vocabulary W. In the same way, we assume that part of speech categories of words in a sentence could also be expressed as $t_1, t_2, \ldots, t_n$, and any part of speech categories could become one of $t_1, t_2, \ldots, t_n$ as long as it is a member of part of speech category set T. With all these assumptions, we define the tagging problem as below.

$$S(w_{1,n}) = \arg\max_{t_{1,n}} P(t_{1,n} \mid w_{1,n}) \tag{1}$$

It means that the most suitable part of speech categories sequence $t_{1,n}$ for sentence $w_{1,n}$ is that with the maximum probability value. We expanded equation (1) into equation (2).

$$\begin{aligned} S(w_{1,n}) &= \arg\max_{t_{1,n}} P(t_{1,n}|w_{1,n}) \\ &= \arg\max_{t_{1,n}} \frac{P(t_{1,n},w_{1,n})}{P(w_{1,n})} \end{aligned} \tag{2}$$

Because the denominator on equation (2) does not have anything to do with any combination of $t_{1,n}$, the maximum value of equation (2) simply depends on its numerator. That makes the equation become as simple as equation (3).

$$S(w_{1,n}) = \arg\max_{t_{1,n}} P(t_{1,n}, w_{1,n}) \tag{3}$$

At this point, we could have simply estimated $P(t_{1,n}, w_{1,n})$ like equation (4) below, where N is the number of sentences on the training data.

$$P(t_{1,n}, w_{1,n}) \approx \frac{C(t_{1,n}, w_{1,n})}{N} \tag{4}$$

However that will not be practical. First, since the training process is carried on the sentence level, it will not create a large number of statistic data. Second, there will not be any universality since each

sentence on the training data will vary in number of words. Because of that, we broke $P(t_{1,n}, w_{1,n})$ on equation (3).

$$P(t_{1,n}, w_{1,n}) = P(t_1 \mid w_1) \prod_{i=2}^{n} P(t_i \mid t_{i-1}, w_i) \tag{5}$$

Next, we substituted $P(t_{1,n}, w_{1,n})$ on equation (3) with what we have got here on equation (5).

$$S(w_{1,n}) = \underset{t_{1,n}}{\arg\max} \left\{ P(t_1 \mid w_1) \prod_{i=2}^{n} P(t_i \mid t_{i-1}, w_i) \right\} \tag{6}$$

Finally, in order to make the process run on word level we modified function $S(w_{1,n})$ into $S'(w_i)$.

$$S'(w_i) = \begin{cases} \underset{t_i \in T}{\arg\max} P(t_i \mid w_i), & \text{for } i = 1 \\ \underset{t_i \in T}{\arg\max} P(t_i \mid t_{i-1}, w_i), & \text{for } 2 \leq i \leq n \end{cases} \tag{7}$$

With this equation we get on with the system.

3 Experiment

3.1 Data and Procedure

First of all, we divide the LOB Corpus into two: the training data and the test data. We do this by excluding a sentence for every ten sentences starting from the second sentence of the corpus. The excluded sentences become the test data and the rest become the training data. The outcomes are a training data with 1,040,475 running words in 49041 sentences and a test data with 116,785 running words in 5449 sentences.

Next, we train the system with the training data. The purpose of the training is to obtain frequency data so that we could approximate the probability function like the equations below.

$$P(t_i = t^j \mid w_i = w^k) \approx \frac{C(t_i = t^j, w_i = w^k)}{C(w_i = w^k)} \tag{8}$$

$$P(t_i = t^j \mid t_{i-1} = t^k, w_i = w^l) \approx \frac{C(t_i = t^j, t_{i-1} = t^k, w_i = w^l)}{C(t_{i-1} = t^k, w_i = w^l)} \tag{9}$$

Where,

- i is an arbitrary natural number on equation (8).
- i is an arbitrary natural number larger than 1 on equation (9).
- $t^j, t^k \in T = \{t^1, t^2, \ldots, t^y\}$
- $w^k, w^l \in W = \{w^1, w^2, \ldots, w^z\}$.

Equation (8) means that the probability of part of speech category t^j become the tag of the word w^k when the word appears is approximated by the appearence frequency of the word with t^j as its tag divided by the appearence frequency of the word w^k.

Almost the same meaning goes to equation (9). The probability of part of speech category t^j become the tag of the word w^l when the previous word's tag is known to be t^k is approximated by the appearence

frequency of the word w^i with t^j as its tag and t^k as its previous word's tag divided by the appearence frequency of the word w^i with t^k as its previous word's tag.

From the training process using equation (8) and (9), we obtain respectively 53,090 combinations of word-tag sequence and 167,156 combinations of tag-word-tag sequence. We then realize that 167,156 combinations were not enough to cover all the possible sequences appearence. We figure that we need at least 7,910,410 (53,090 X 149) combinations (the figure 149 was from the number of part of speech categories used in this experiment) to cover all the possible sequences appearence. Therefore, we modify equation (7) into equation(10), where α is the weight given to the first term of the equation.

$$S'(w_i) = \arg \max_{t_i \in T} \{\alpha P(t_i \mid t_{i-1}, w_i) + (1-\alpha) P(t_i \mid w_i)\} \tag{10}$$

With this equation we carry several experiments.

3.2 Weight α

With the α is fixed at 0.8, we test the system on the test data. And the result is, out of 116,785 running words on the test data, 111,613 of them are correctly tagged. That is an accuracy of 95.571%.

In the second experiment, we try to do an optimization. The optimization is done by fixing the value of weight α at every beginning of a sentence. The equation for fixing α is as follows, where f is the total number of words which has a non-zero value of $\alpha P(t_i \mid t_{i-1}, w_i)$ so far, and s is the total number of running words which was processed so far. With this optimization, we hope that value of weight α is always reflecting the ratio between word belongs to first order Markov Model and word belongs to second order Markov Model at any given time.

$$\alpha = \frac{f}{s} \tag{11}$$

The result is better than the first experiment. Out of 116,785 words, 111,629 are correctly tagged and that means the accuracy is improved to 95.585%.

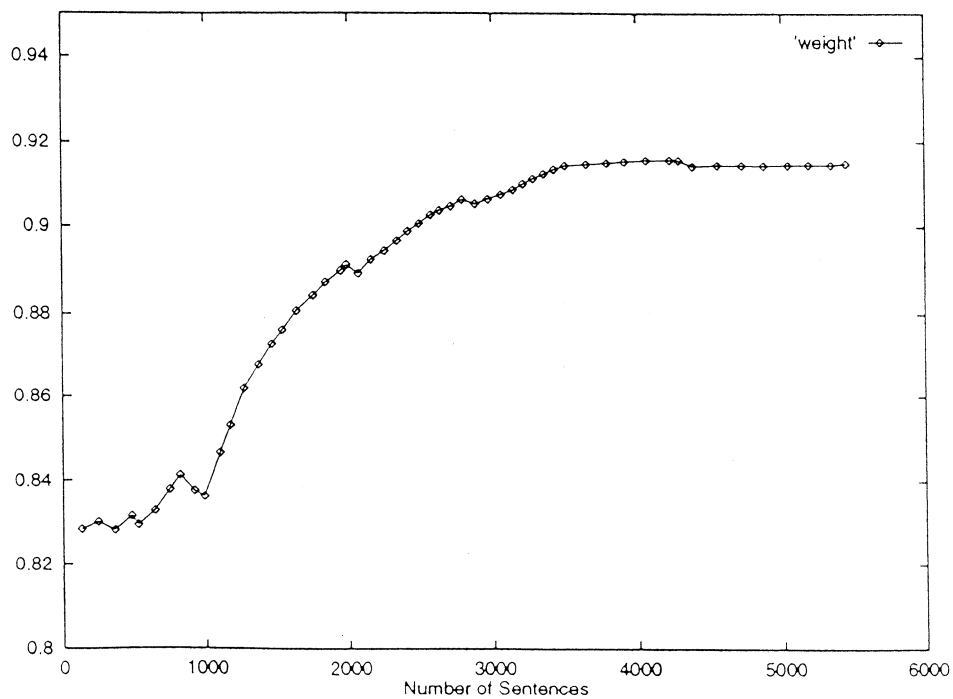


Fig.1 Weight α's Behaviour During Experiment

We investigate α's behaviour during experiment. We take α's value at every hundred sentences. The result is shown on fig.1. From this investigation we also know that out of 116,785 words on the test data, 106,837 of them are tagged by the first term of equation (10). And that makes α converged on 0.914818 (106,837/116,785). We then carry the third experiment with weight α was fixed at 0.914818. The result is 111,630 words are correctly tagged, an accuracy of 95.586%.

4 Conclusions

The high degree of lexical category ambiguity in languages such as English poses problems for parsing. A system which provides an output of disambiguated words will be a great help to the parsing task.

In this paper, we proposed a method to disambiguate words in a sentence using statistic data which was derived from a training data. The derived data was in form of part of speech category - word combination, which has the properties of the possible part of speech category and its probability. We carried several experiments with this model and our system at its best performance reached an accuracy of 95.586%.

We consider that this model still opens a wide possibility for modification. One of the future homework is modification on the statistic data's form. Besides the part of speech category combination which was used by us in this paper, there are still other type of combinations possible for exploration.

On the other hand, our proposed model took a lot of time for the training process. Also, a huge amount of training data was required. This inconvinience of preparing a large data and training it in a quite long time should be taken into consideration when we evaluate a tagging system. In the near future, we would like to propose a new evaluation method of a tagging system which at least takes the ratio between the training data and the test data into its consideration.

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